

(P₂)

A	P(A)
T	0.70
F	0.01

date: 17/07/2026
day: Friday.

A company wants to classify whether a customer will buy a computer based on their age

customer	age	Buy Computer
1	young	yes
2	young	yes
3	middle	yes
4	old	no
5	old	no
6	old	yes
7	middle	no
8	young	no
9	young	yes
10	old	yes

Question:
using the Naïve Bayes classifier, Predict whether
a customer with Age - young will buy a computer

Problem - 2

A hospital wants to predict whether a patient
has diabetes based on continuous medical
measurements.

Patient	Glucose	BMI	Diabetes
		22.5	NO
P ₁	85	24.0	NO
P ₂	90	23.8	NO
P ₃	95	30.5	yes
P ₄	120	32.0	yes
P ₅	135	35.0	yes
P ₆	150		

New Patient:

Glucose

130

BMI

31

Question:
using the gaussian naïve Bayes
classifier, Predict whether the new
has diabetes

What's
computer

Problem - 1

Age

yes

no

young

middle

old

using

Multinomial Method

Total Customers = 10

Buy $\Rightarrow$ yes $\Rightarrow 6$

Buy $\Rightarrow$ no $\Rightarrow 4$

Calculate Prior Probabilities

$$P(\text{yes}) = 6/10 = 0.6$$

$$P(\text{no}) = 4/10 = 0.4$$

Calculate Likelihood

Buy $\Rightarrow$ yes

$$P(\text{young}|\text{yes}) = 3/6 = 0.5$$

Buy $\Rightarrow$ no

$$P(\text{young}|\text{no}) = 1/4 = 0.25$$

Posterior Probability.

yes,

$$P(\text{yes}) \times P(\text{young}|\text{yes}) = 0.6 \times 0.5 = 0.3$$

for, no

$$P(\text{no}) \times P(\text{young}|\text{no}) = 0.4 \times 0.25 = 0.1$$

if yes $\Rightarrow 0.30$, no $\Rightarrow 0.10$
since $0.80 > 0.10$, the prediction is
Buy computer $\Rightarrow$ yes

Problem 2:

New Patient

Glucose $\Rightarrow 130 \text{ mg/dL}$

BMI $\Rightarrow 31 \text{ kg/m}^2$

Separate the data

NO Glucose $\Rightarrow 85, 90, 95$

BMI $\Rightarrow 28.5, 24.0, 23.8$

yes

Glucose $\Rightarrow 120, 135, 150$

BMI $\Rightarrow 30.5, 22.0, 35.0$

mean;

yes, no

mean glucose $\Rightarrow \frac{85+90+95}{3} \Rightarrow 90$

BMI

$$\mu \Rightarrow \frac{22.5 + 24 + 23.8}{3} \Rightarrow 23.43$$

for, yes

$$\mu \Rightarrow \frac{120 + 135 + 150}{3} \Rightarrow 135$$

mean, BMI

$$\mu \Rightarrow \frac{30.5 + 32 + 35}{3} \Rightarrow 32.5$$

compare New Patient

New Glucose $\Rightarrow 120$

close to 135 (yes)

Far from 90 (No)

New BMI $\Rightarrow 31$

Close to 32.5 (yes)

Far from 23.43 (No)

$\therefore$ the Gaussian Probability for the
Yes class will be much higher

Diabetes $\Rightarrow$ yes

20/50 (TP)